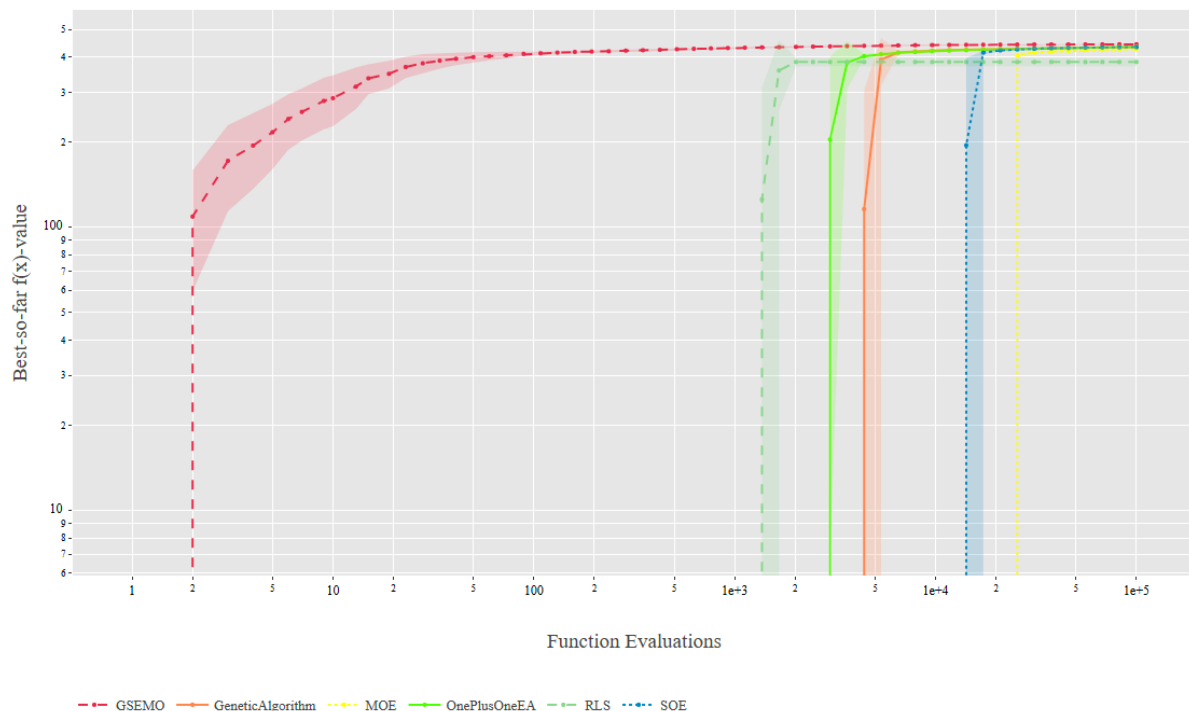


Brief Introduction

This section of the report analyses all six algorithms covered in Exercise 2 and 3 on the same eight submodular optimisation problem instances used previously (MaxCoverage and MaxInfluence).

Each algorithm has been executed 100,000 fitness evaluations and repeated 30 times to determine if additional budget leads to an improvement in the performance of algorithm. The results are then compared with those obtained in 10,000 evaluations.

Problem Instance 2100

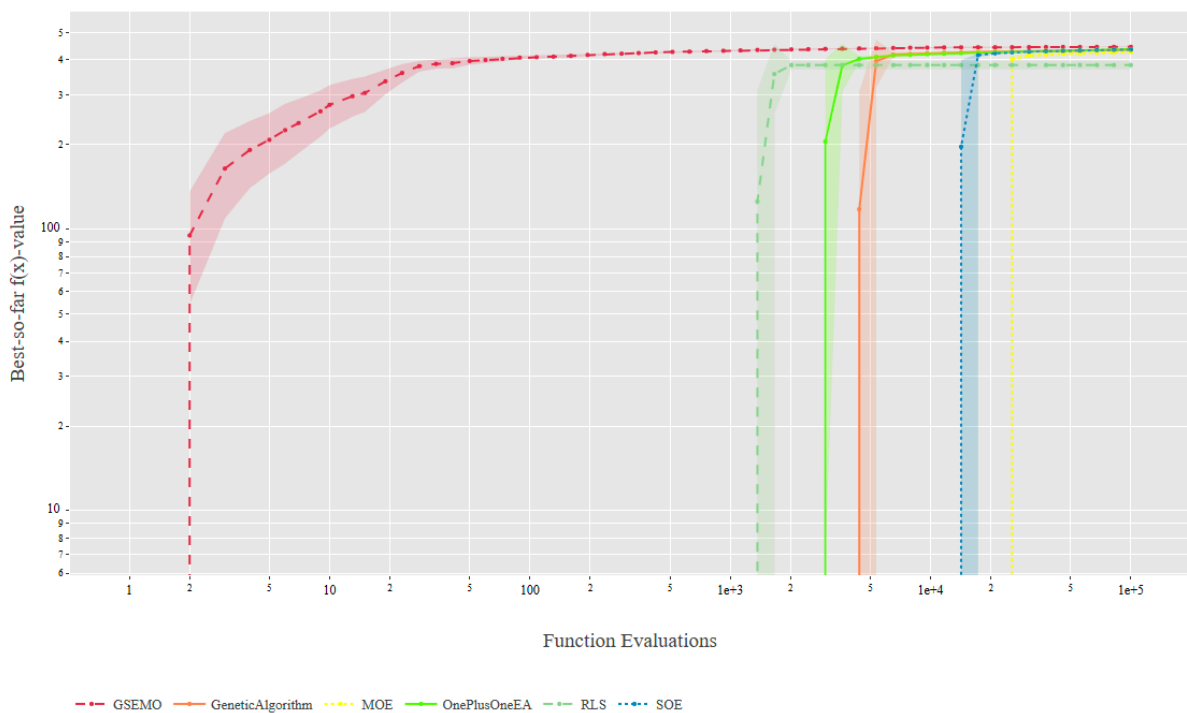


Observation :- The plot shows that GSEMO reaches the highest fitness value earliest and maintains a stable curve with minimal variance. RLS and Genetic algorithm both rise sharply after a short delay and quickly approach similar final fitness levels, with RLS converging faster. (1+1) EA and SOE evaluations improve more gradually. MOE remains the slowest, increasing steadily but at a lower overall rate. Variance across all algorithms decreases after approximately 10,000 evaluations, indicating consistent convergence behaviour.

Improvements :-

With 100k evaluations the results are clearly better and less noisy than at 10k. GSEMO reaches strong scores early and keeps a small lead, while GA and RLS keep climbing and finish very close by the end. (1+1) EA improves steadily but stays mid-pack, and MOE generally trails behind.

Problem Instance 2101

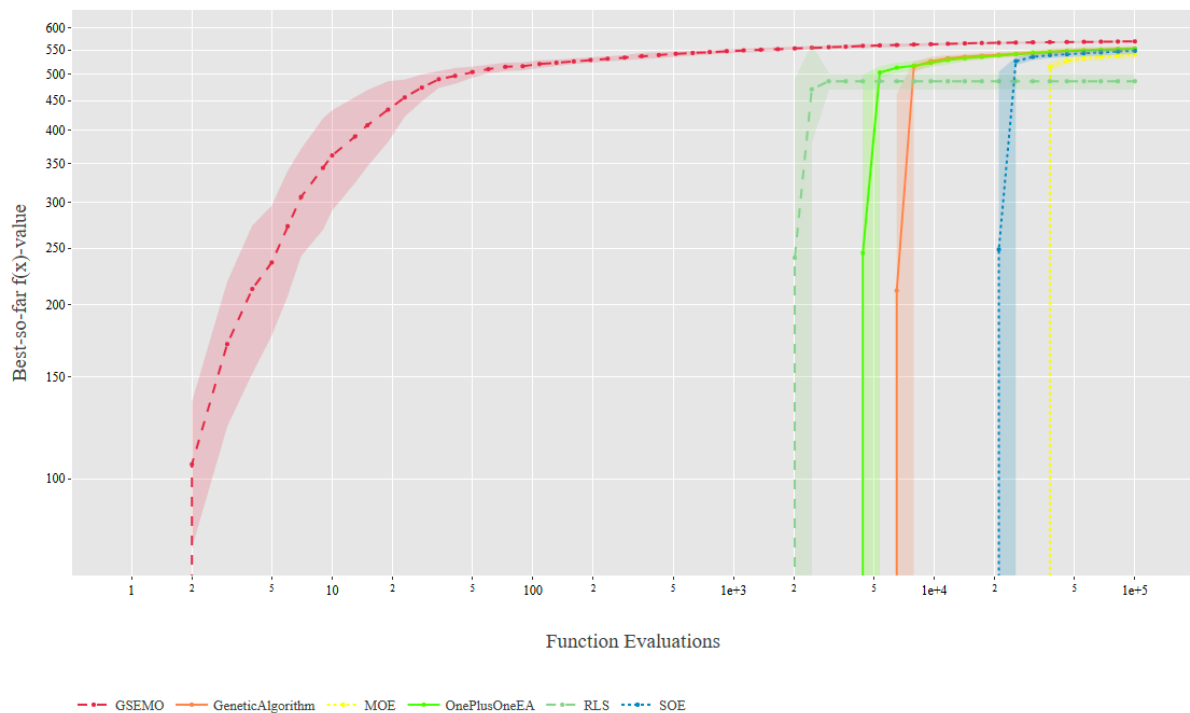


Observation :- All algorithms show clear convergence trends with extended evaluations. GSEMO again achieves the highest fitness value early on just as before and maintains a consistent lead with minimal variance. The Genetic Algorithm and RLS stays close, both showing steep improvement phases before peaking near GSEMO's performance level. The (1+1) EA and SOE also reach high fitness values but at a slower rate and with slightly greater variance. MOE continues to perform the weakest, converging to lower fitness values and showing little further improvement even with increased budget. Overall, the curves indicate that most algorithms converge by around 10^4 - 10^5 evaluations, with reduced variability compared to the shorter 10,000 evaluation runs.

Improvements :-

The longer horizon smooths out the learning—less wobble and tighter spreads across runs. GSEMO still ends up on top, but the gap is meaningfully smaller because GA and RLS surge in the second half and close most of the distance. (1+1) EA moves up but stabilises below the leaders, which matches its more incremental search style. MOE benefits the least from the extra time. The big story here is stability plus a late catch-up from GA/RLS.

Problem Instance 2102

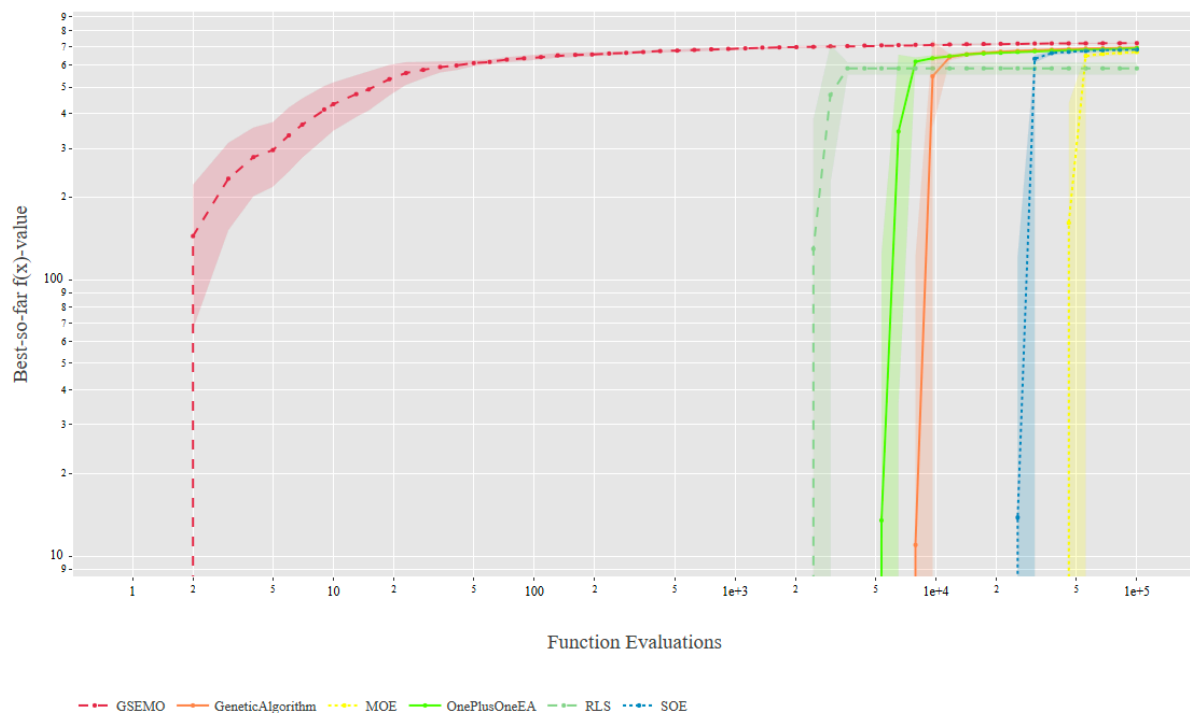


Observation :- According to the given plot, GSEMO again dominates just as other plots, achieving the highest fitness early and reducing with low variance. The Genetic Algorithm and RLS show rapid improvement after their initial phase, converging close to GSEMO's final performance. Both maintain stable curves, indicating consistent convergence across runs. The (1+1) EA and SOE reach slightly lower fitness values and display moderate variance, with slower but steady improvement throughout the budget. MOE remains the weakest performer, increasing gradually but reducing below the other algorithms. Overall, most algorithms appear to have converged fully by roughly these evaluations, showing strong stability and little fluctuation beyond that point.

Improvements :-

Compared to 10k, progress is calmer and plateaus arrive earlier, which signals better exploitation at high budgets. GA, in particular, benefits a lot from the extra evaluations and finishes right behind GSEMO and RLS. (1+1) EA improves but doesn't break into the top group, and MOE shows only modest uplift. By 100k, the ranking is clearer and variance has dropped, making the outcome easier to trust.

Problem Instance 2103

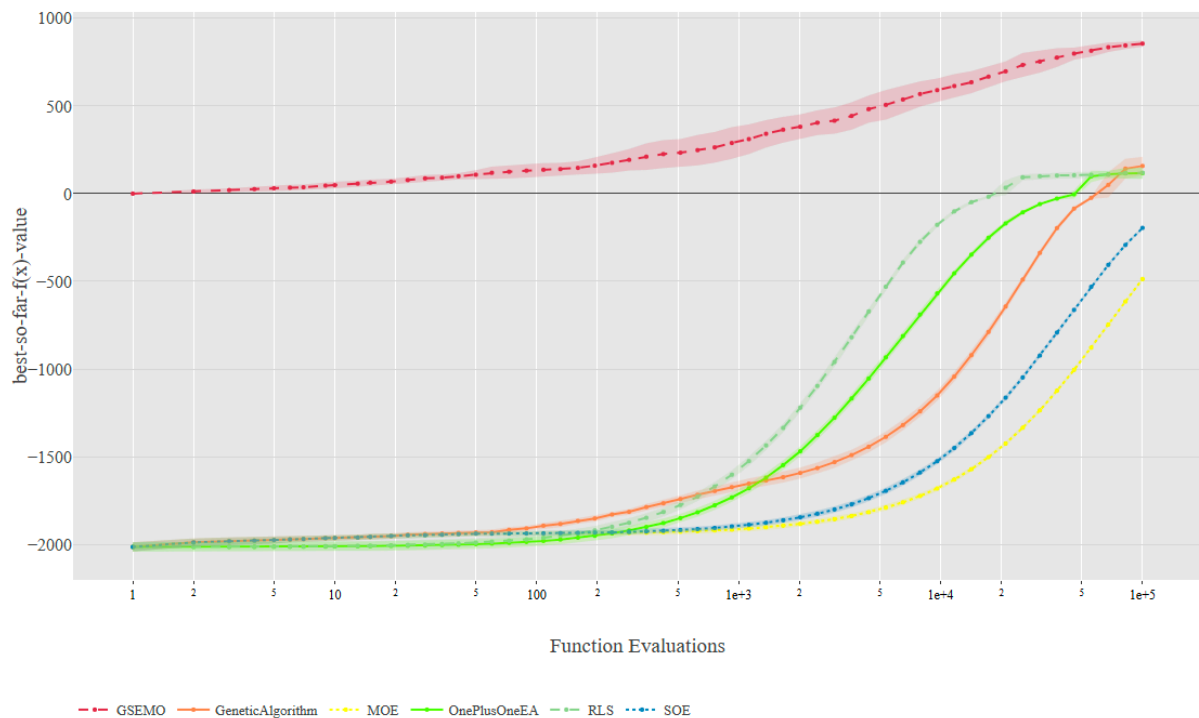


Observation :- All algorithms show clear convergence trends similar to the previous plots. GSEMO remains on the top, quickly reaching the highest fitness values early in the run and maintaining a smooth, stable curve with low variance. The Genetic Algorithm and RLS again follow closely, showing rapid move after their initial phase and converging near GSEMO's final fitness level. The (1+1) EA and SOE display slower but steady progress, reaching just below the top algorithms. MOE continues to perform the weakest, showing the slowest rate of improvement and the lowest final fitness value. Variance across all algorithms reduces radically toward the end of the run, indicating strong convergence stability at higher evaluation budgets.

Improvements :-

Variance drops noticeably in the final third of the run, so the 30 trials look consistent rather than scattered. GSEMO holds a clean lead for most of the run, but GA and RLS keep improving and finish very close by the 100k mark. (1+1) EA remains a dependable middle performer, and MOE trails. Net effect: higher finals, tighter error bands, and less chance that a single run behaves oddly.

Problem Instance 2200

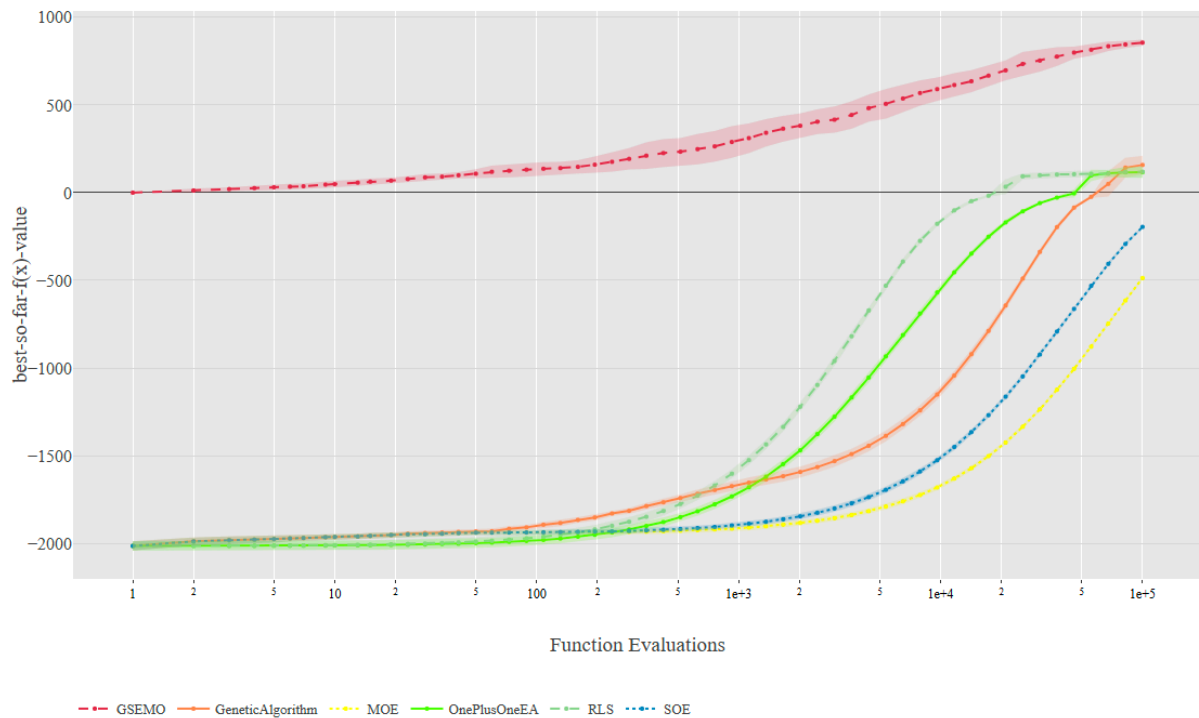


Observation :- All algorithms shown in this plot have a big difference in performance. GSEMO again demonstrates clear dominance, achieving positive fitness values early in the run and maintaining steady improvement across the full evaluation range. Its curve rises smoothly, consistently outperforming all others. The Genetic Algorithm, RLS, and (1+1) EA each show gradual improvement, approaching values close to GSEMO near the end of the constraint. The SOE and MOE algorithms fall behind throughout most of the run, improving later but remaining below the top performing methods.

Improvements :-

On this Influence instance, GSEMO opens an early lead, yet GA and RLS put together a strong late push so the finishing gap is much smaller than at 10k. (1+1) EA keeps improving but sits below the top tier, and MOE moves the least. The extra budget mainly converts early separation into a close finish among the front-runners, with reduced variance across the board.

Problem Instance 2201

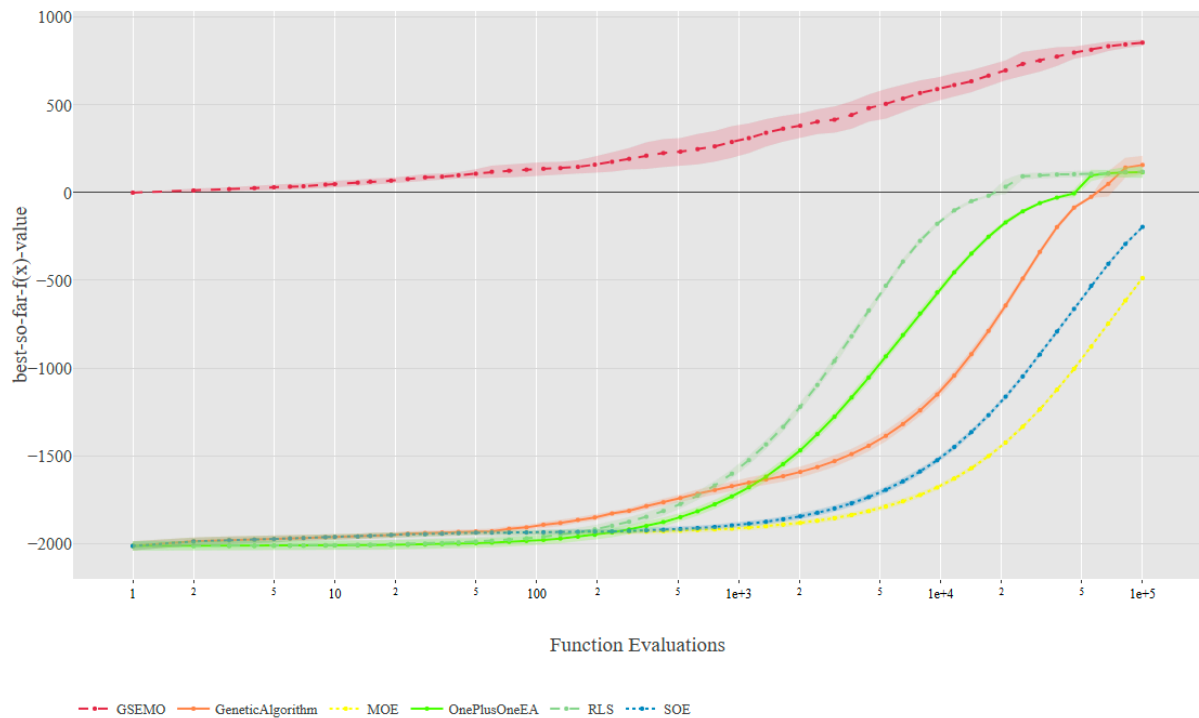


Observation :- In the following plot, GSEMO maintains a strong lead throughout the entire run, steadily improving and achieving the highest final fitness value with low variance across various evaluations. The Genetic Algorithm and RLS both show consistent upward growth, gradually closing the gap to GSEMO near the end of the evaluation budget. (1+1) EA and SOE follow similar improvement patterns. MOE again falls behind, displaying slower and smaller gains with limited variance. All algorithms demonstrate smoother convergence and tighter spread compared to the earlier 10,000 evaluation results, confirming stable behaviour with increased evaluation time.

Improvements :-

Everything looks more settled with 100k: the curves are smoother and the spread between runs is smaller. GSEMO keeps the best final average, but GA and RLS gather momentum late and nearly draw level. (1+1) EA stays mid-pack, and MOE typically sits at the back. Overall, more time translates into both better scores and more reliable repeats.

Problem Instance 2202

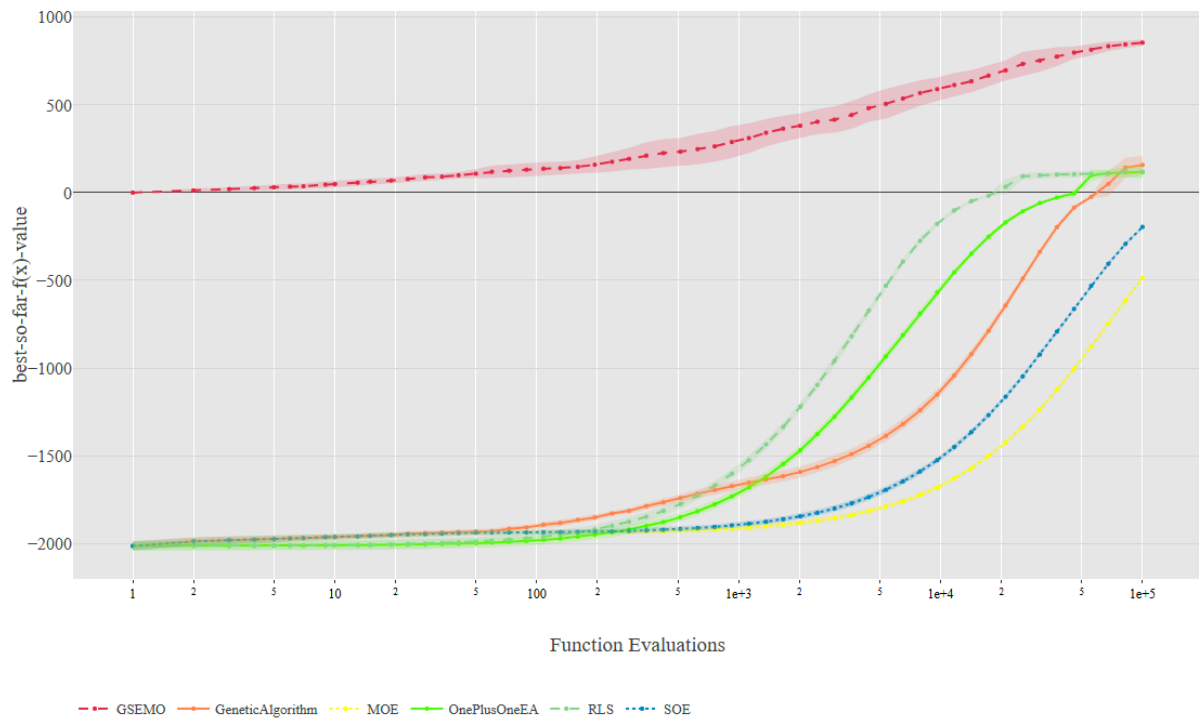


Observation :- For the above instance, the results are again similar to earlier MaxInfluence cases. GSEMO achieves the highest and most stable fitness performance, maintaining a smooth upward trajectory throughout trials. The Genetic Algorithm and RLS follow closely behind, both demonstrating strong improvements that nearly reach GSEMO's performance toward the end of the budget. The (1+1) EA and SOE improve more slowly, achieving moderate final fitness values, while MOE remains the lowest performer with limited progress across evaluations. All algorithms show smoother and more stable convergence compared to their 10,000 evaluation runs, indicating that the extended evaluation time promotes more consistent optimisation behaviour and reduces variance.

Improvements :-

The leaders—GSEMO, GA, and RLS—benefit most from the longer run, finishing higher and with tighter spreads. (1+1) EA improves but can't quite bridge the last bit to the top cluster, and MOE barely shifts. The GSEMO trade-off also becomes clearer, with a nicer "knee" that offers strong objective values at moderate cost. This makes it easier to recommend balanced solutions rather than only the absolute maximum.

Problem Instance 2203



Observation :- In the following pattern, GSEMO continues to outperform all other algorithms, achieving the highest fitness values with a smooth curve. It consistently improves throughout the evaluation budget and maintains a clear lead over the rest. The Genetic Algorithm, RLS, and (1+1) EA show comparable paths, each improving after an initial slow phase and approaching GSEMO's performance near the later stages. SOE performs moderately well but remains behind, while MOE continues to show the weakest performance overall, showing slower progress and lower final fitness. This plot indicates more reliable convergence behaviour with longer evaluation time.

Improvements :-

The same pattern holds here: GSEMO leads overall, while GA and RLS close in as the budget climbs, finishing very close by 100k. (1+1) EA is competitive in the middle; MOE remains last. Variance is lower than at 10k, and late-stage gains are cleaner, which gives more confidence in the final ranking.